

Problem set #2- Supervised Learning II

2. a) For logistic regression, the loss function is given by

$$L(\theta) = \sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))$$

After training: $\frac{dL(\theta)}{d\theta} = 0$

$$\sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)} = 0$$

set $j=0$, $x_0^{(i)} = 1$

$$\sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)})) = 0$$

$$\sum_{i=1}^m h_{\theta}(x^{(i)}) = \sum_{i=1}^m y^{(i)}$$

$$h_{\theta}(x^{(i)}) = P(y^{(i)}=1 | x^{(i)}, \theta), \quad y^{(i)} = I\{y^{(i)}=1\}$$

$$\sum_{i=1}^m P(y^{(i)}=1 | x^{(i)}, \theta) = \sum_{i=1}^m I\{y^{(i)}=1\}$$

Divide both $| \{i \in I_{a,b}\} |$
where $(a,b) = (0,1)$

$$\frac{\sum_{i=1}^m P(y^{(i)}=1 | x^{(i)}, \theta)}{| \{i \in I_{a,b}\} |} = \frac{\sum_{i=1}^m I\{y^{(i)}=1\}}{| \{i \in I_{a,b}\} |}$$

(b) Even if model is perfectly calibrated, it cannot be trained to achieve perfect accuracy (100% accuracy)

(c) By adding regularization, ~~the loss~~ the loss function becomes

$$L(\theta) = \sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) + \frac{\lambda}{2} \|\theta_x\|^2$$

After training, the ~~gr~~ gradients become 0.

$$\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} + \lambda \theta_j = 0$$

Let $j=0$, $x_0^{(i)} = 1$

$$\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) + \lambda \theta_0$$

$$\sum_{i=1}^m h_{\theta}(x^{(i)}) + \lambda \theta_0 = \sum_{i=1}^m y^{(i)}$$

$$\sum_{i=1}^m P(y^{(i)} = 1 | x^{(i)}; \theta) + \lambda \theta_0 = \sum_{i=1}^m I\{y^{(i)} = 1\}$$

The model is not perfectly calibrated, as ^{for} a perfectly calibrated model,

$$\frac{\sum P(y^{(i)} = 1 | x^{(i)}; \theta)}{|\{i \in I_{a,b}\}|} = \frac{\sum I\{y^{(i)} = 1\}}{|\{i \in I_{a,b}\}|}$$

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Q3. a) Bayesian Interpretation of Regularization Background

we know, $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$$p(\theta|x,y) = \frac{p(x,y,\theta)}{p(x,y)} = \frac{p(y|x,\theta) \cdot p(x,\theta)}{p(x,y)}$$

$$= \frac{p(y|x,\theta) p(\theta|x) p(x)}{p(x,y)}$$

Acc. to the question, $p(\theta) = p(\theta|x)$

so

$$p(\theta|x,y) = \frac{p(y|x,\theta) p(\theta|x) \cdot p(x)}{p(x,y)}$$

$$= \frac{p(y|x,\theta) \cdot p(\theta) p(x)}{p(x,y)}$$

$$\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta|x,y) = \arg \max_{\theta} p(y|x,\theta)$$

$$\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta|x,y) = \arg \max_{\theta} p(y|x,\theta) p(\theta) \cdot \frac{p(x)}{p(x,y)}$$

$$= \arg \max_{\theta} p(y|x,\theta) \cdot p(\theta)$$

since $\frac{p(x)}{p(x,y)}$ is a constant.

$$b) \theta_{MAP} = \arg \max_{\theta} p(y|x, \theta) \cdot p(\theta)$$

$$= \arg \max_{\theta} \log (p(y|x, \theta) p(\theta))$$

$$= \arg \max_{\theta} \log p(y|x, \theta) + \log p(\theta)$$

$$= \arg \min_{\theta} -\log p(y|x, \theta) - \log p(\theta)$$

$$\theta \sim \mathcal{N}(0, \eta^2 I)$$

$$p(\theta) = \frac{1}{(2\pi)^{n/2} \eta^n} \exp \left\{ -\frac{1}{2\eta^2} \theta^T \theta \right\}$$

$$p(\theta) = (2\pi)^{-n/2} \eta^{-n} \exp \left\{ -\frac{1}{2\eta^2} \|\theta\|^2 \right\}$$

$$\log p(\theta) = -\frac{n}{2} \log(2\pi) - n \log \eta - \frac{1}{2\eta^2} \|\theta\|^2$$

$$\theta_{MAP} = \arg \min_{\theta} -\log p(y|x, \theta) - \log p(\theta)$$

$$= \arg \min_{\theta} -\log p(y|x, \theta) + \frac{1}{2\eta^2} \|\theta\|^2$$

$$\lambda = \frac{1}{2\eta^2}$$

d) Laplace distribution

$$\theta \sim L(0, bI)$$

$$p(\theta) = \frac{1}{(2b)^n} \exp \left\{ -\frac{1}{b} \|\theta\|_1 \right\}$$

$$\log p(\theta) = -n \log(2b) - \frac{1}{b} \|\theta\|_1$$

$$\theta_{\text{MAP}} = \arg \min_{\theta} \frac{1}{2\sigma^2} \|\mathbf{x}\theta - \vec{y}\|^2 - \log(p(\theta))$$

$$= \arg \min_{\theta} \frac{1}{2\sigma^2} \|\mathbf{x}\theta - \vec{y}\|^2 + \frac{1}{b} \|\theta\|_1$$

From the question

$$J(\theta) = \|\mathbf{x}\theta - \vec{y}\|^2 + \gamma \|\theta\|_1$$

$$\theta_{\text{MAP}} = \arg \min_{\theta} J(\theta)$$

$$\boxed{\gamma = \frac{2\sigma^2}{b}}$$

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$K_1, K_2 \rightarrow$ kernels over $\mathbb{R}^n \times \mathbb{R}^n$, $a \in \mathbb{R}^+$ be a positive real number.

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a real valued function

$\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ be a function mapping from $\mathbb{R}^n$ to $\mathbb{R}^d$.

a) $K(x, z) = K_1(x, z) + K_2(x, z)$

since both K_1, K_2 are both positive semi-definite, their addition would also be positive semi-definite.

b) $K(x, z) = K_1(x, z) - K_2(x, z)$

It might be possible that $K_2(x, z) > K_1(x, z)$, so in that case $K(x, z)$ would not be positive semi-definite.

c) $K(x, z) = a K_1(x, z)$

Since a is a positive real number, & $K_1(x, z)$ is positive semi-definite, so $K(x, z)$ is positive semi-definite.

d) $K(x, z) = -a K_1(x, z)$

since a is a positive real number, & $K_1(x, z)$ is positive semi-definite, so $K(x, z)$ is ^{not} positive semi-definite.

e) $K(x, z) = K_1(x, z) \cdot K_2(x, z)$

Since K_1, K_2 are both positive semi definite kernels, their product would also be positive semi definite.

f) $K(x, z) = f(x) \cdot f(z)$

For any vector z ,

$$z^T K z = \sum_i \sum_j z_i K_{ij} z_j$$

$$= \sum_i \sum_j z_i \phi(x^{(i)})^T \phi(x^{(j)})^T z_j$$

$$= \sum_i \sum_j z_i \sum_K \phi(x^{(i)})_K \phi(x^{(j)})_K z_j$$

$$= \sum_K \sum_i \sum_j z_i (\phi(x^{(i)})_K \phi(x^{(j)})_K z_j$$

$$= \sum_K \left(\sum_i z_i \phi(x^{(i)})_K \right)^2 \geq 0, \text{ hence it is a kernel as it positive semi definite.}$$

g) $K(x, z) = K_3(\phi(x), \phi(z))$

$K(x, z)$ would be kernel as $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ & K_3 is also a kernel as per the problem statement.

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Q5) Kernelizing the Perceptron

a) i) Binary classification $y \in \{0, \pm 1\}$

$$h_{\theta}(x) = g(\theta^T x) \quad g(z) = \text{sign}(z) = \pm 1 \text{ if } z \geq 0$$

$$\theta^{(i+1)} = \theta^{(i)} + \alpha (y^{(i+1)} - h_{\theta^{(i)}}(x^{(i+1)})) x^{(i+1)}$$

$$\theta^{(i+1)} = \sum_{j=1}^i \beta_j \phi(x^{(j)}), \theta^{(0)} = \sum_{j=1}^0 \beta_j \phi(x^{(j)}) = \vec{0}$$

$$\begin{aligned} \text{(ii)} \quad h_{\theta^{(i)}}(x^{(i+1)}) &= g(\theta^{(i)T} \phi(x^{(i+1)})) \\ &= \text{sign}(\theta^{(i)T} \phi(x^{(i+1)})) \\ &= \text{sign}\left(\sum_{j=1}^i \beta_j \phi(x^{(j)})^T \phi(x^{(i+1)})\right) \\ &= \text{sign}\left(\sum_{j=1}^i \beta_j (\phi(x^{(j)}), \phi(x^{(i+1)}))\right) \\ &= \text{sign}\left(\sum_{j=1}^i \beta_j K(x^{(j)}, x^{(i+1)})\right) \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \theta^{(i+1)} &= \theta^{(i)} + \alpha (y^{(i+1)} - h_{\theta^{(i)}}(x^{(i+1)})) x^{(i+1)} \\ \theta^{(i+1)} &= \theta^{(i)} + \alpha (y^{(i+1)} - \text{sign}(\sum_{j=1}^i \beta_j K(x^{(j)}, x^{(i+1)}))) \phi(x^{(i+1)}) \\ &= \sum_{j=1}^i \beta_j \phi(x^{(j)}) + \alpha (y^{(i+1)} - \text{sign}(\sum_{j=1}^i \beta_j K(x^{(j)}, x^{(i+1)}))) \phi(x^{(i+1)}) \\ \beta_{i+1} &= \alpha (y^{(i+1)} - \text{sign}(\sum_{j=1}^{i+1} \beta_j K(x^{(j)}, x^{(i+1)}))) \end{aligned}$$